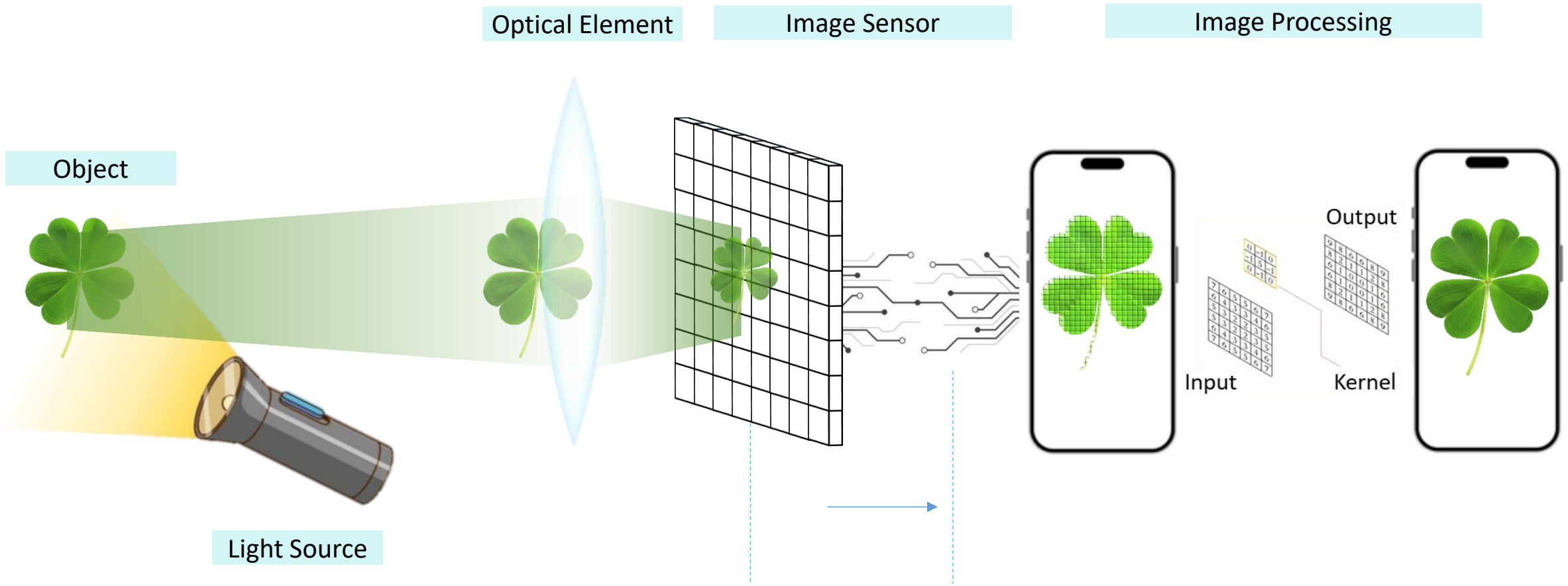
Research Experience Summary

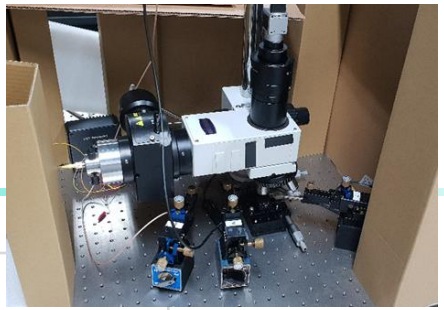
2026.03

Personal Background



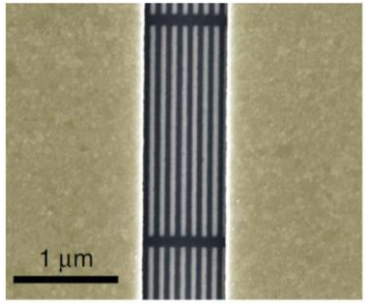
My research interest is to pave novel visual information beyond the Visible
based on Electromagnetics and Optoelectronics.

Experience ① Graphene Based Optoelectronic Devices



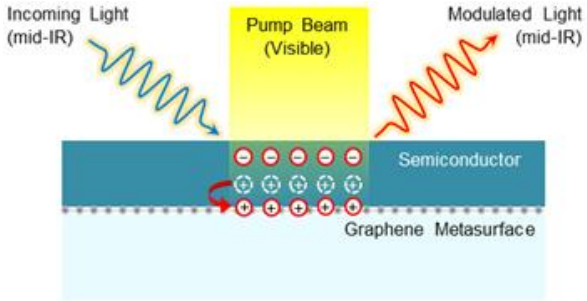
Evaluation System

Graphene Plasmonics

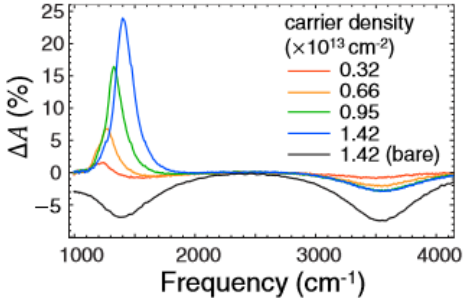


Plasmonic Frequency in Graphene Metasurface

$$\hbar\omega_p = 4e\sqrt{\frac{1}{\pi}\frac{2}{\varepsilon_1+\varepsilon_2}}\sqrt{\frac{E_F}{W}}$$



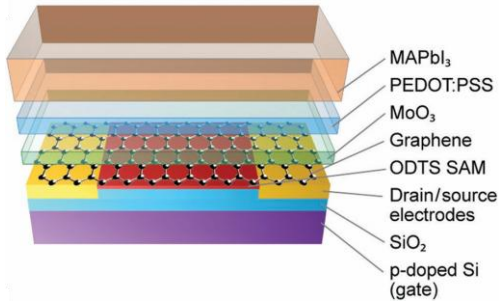
Concept of Full Optical Modulator



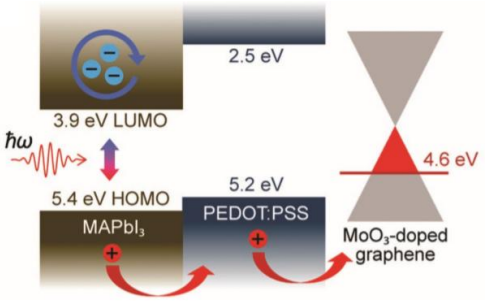
Variation of Plasmonic Frequency

Jang Min Seok et al., *Physical Review B,* 2014, 90.16, 165409.

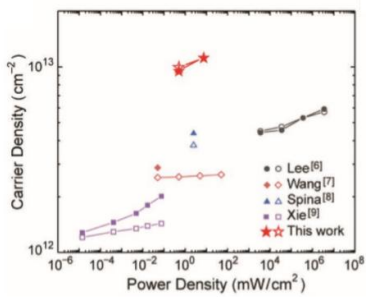
Photodoping of Graphene



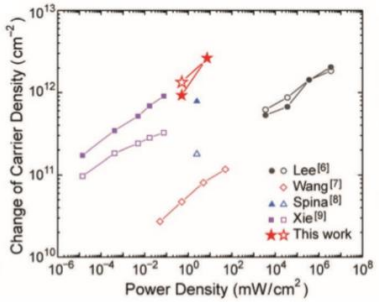
Optoelectronic Device Structure



Concept of Photodoping



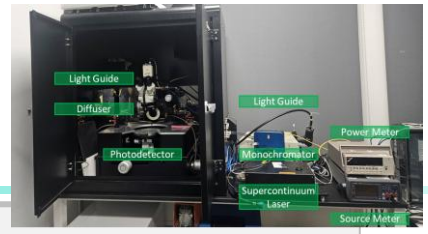
Carrier Density before Doping



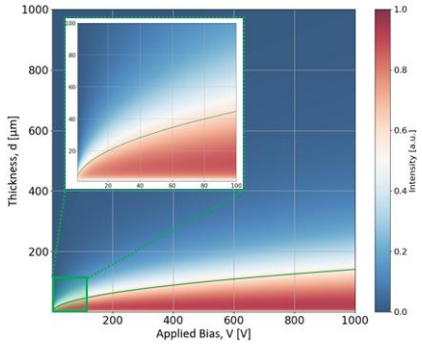
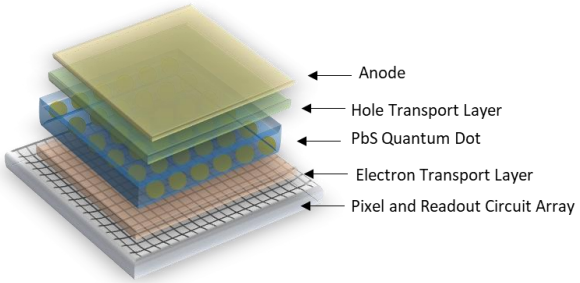
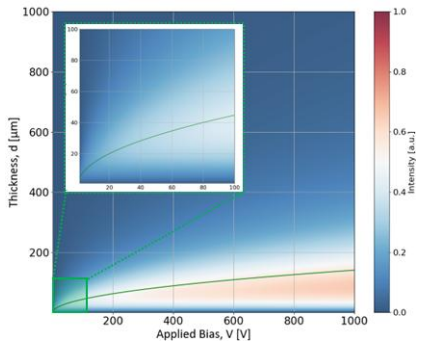
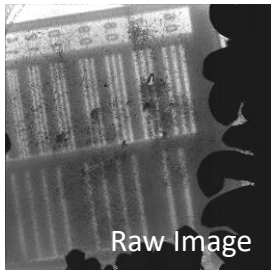
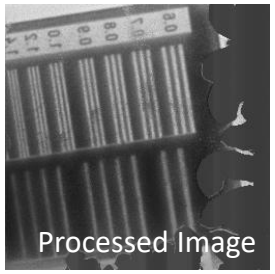
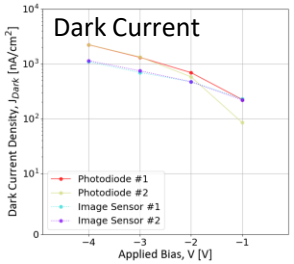
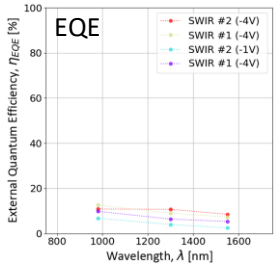
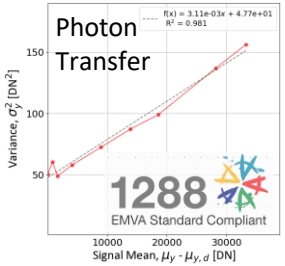
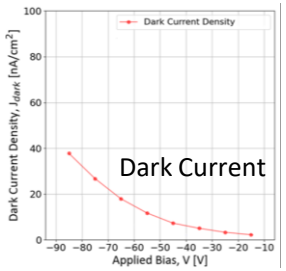
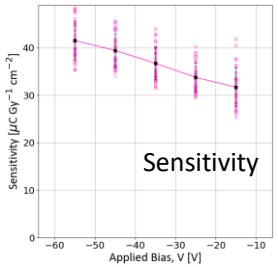
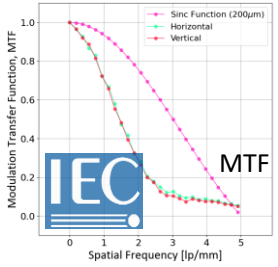
Carrier Density after Doping

S Kim et al., "Efficient Photodoping of Graphene in Perovskite–Graphene Heterostructure" , Advanced Electronic Materials, 2019.

Experience ② Monolithic Image Sensor



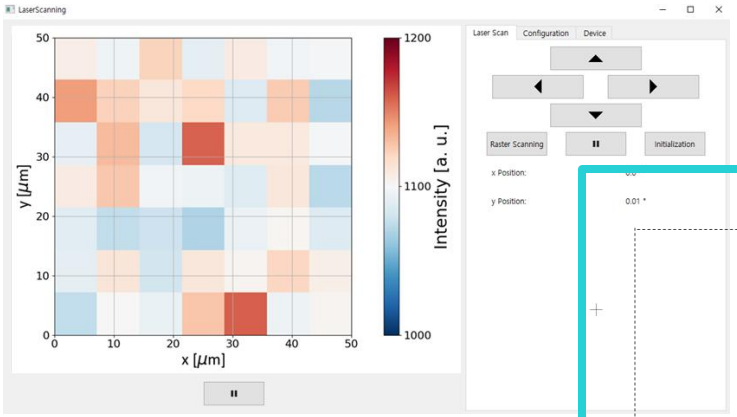
Evaluation System

	SWIR	X-Ray*
Photodetector Design	 	 For Drift Dominant Case
Modelling Function	Charge Collection Efficiency, $\eta_{CCE} = \frac{DOS(\lambda)E_{in}(\lambda)}{E_g} \times (1 - e^{-k_0 \kappa x}) * \frac{\mu\tau E}{d-x}\left(1 - e^{-\frac{d-x}{\mu\tau E}}\right)$	
Image	Image Sensor Raw Image Processed Image	Image Sensor  Raw Image  Processed Image
Figure of Merit	 Dark Current  EQE  Photon Transfer	 Dark Current  Sensitivity  MTF

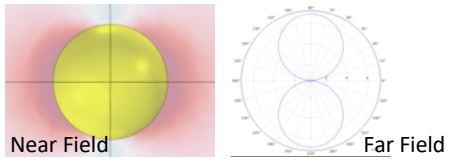
* S Kim et al., "Quantum Dot-Based Direct X-Ray Detector for Mammography", SPIE Medical Imaging (2026): 13024-49

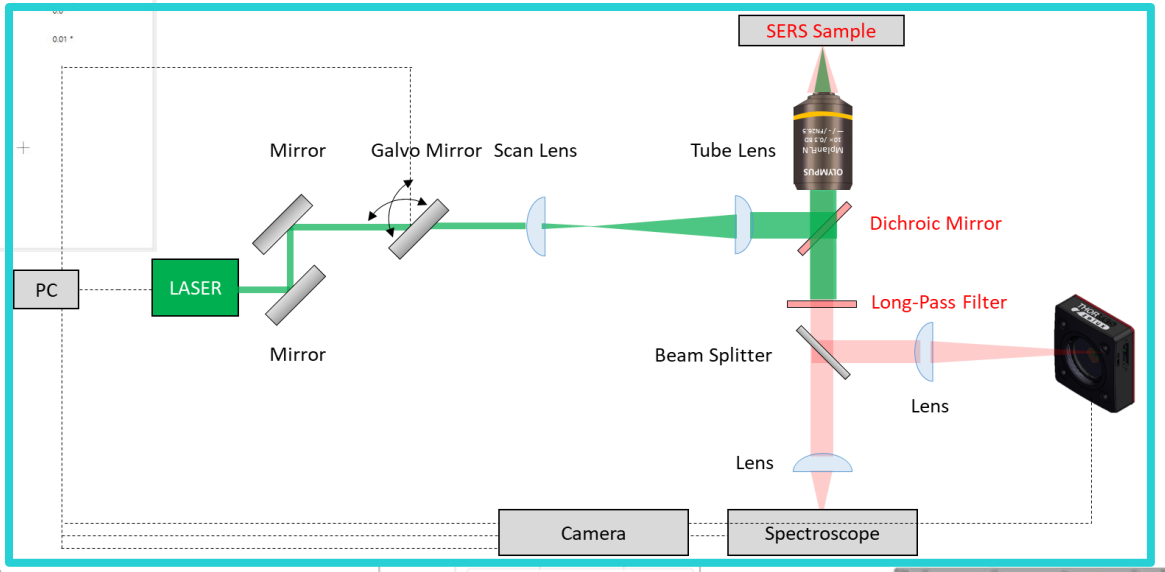
Experience ③ Laser Scanning Raman Spectroscope

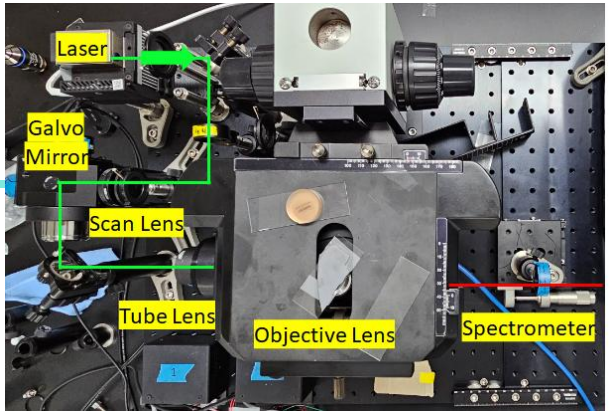
Beam Scanning GUI



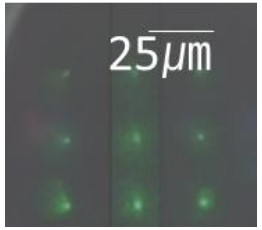
SERS Analysis*, **



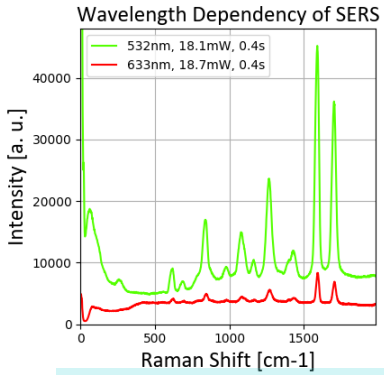


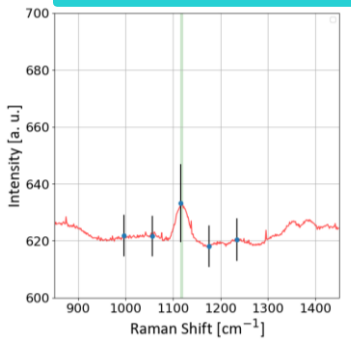


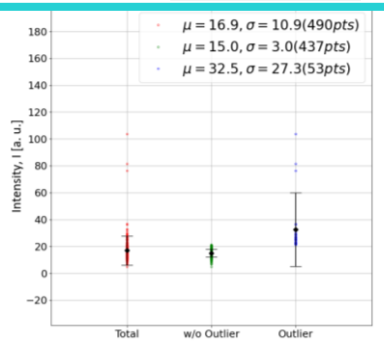
Raman Spectroscope Set-up



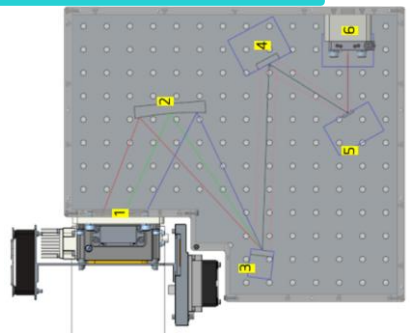
Beam Scanning Monitoring

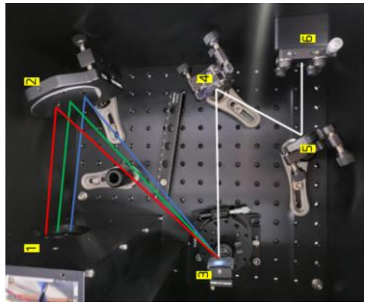






Results and Data Analysis



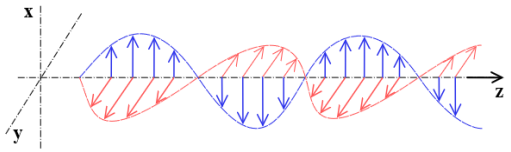


Spectroscope Set-up

*, S Kim, K Yoon, "Method and device for detecting surface-enhanced raman scattering ", KR20240115636A
** S Kim, K Yoon, "Method and apparatus for measuring surface-enhanced raman scattering", WO2025170222A1

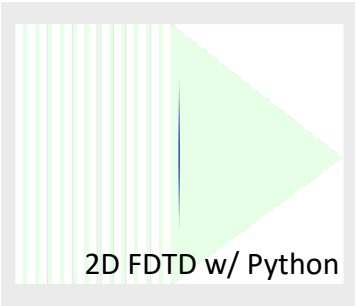
Personal Interests: Electromagnetic-Matter Interaction

Electromagnetic Wave



$$\mathbf{E} = \mathrm{E}_0 \bar{\mathbf{a}}_\mathbf{r} e^{-j(\omega t - kr)}$$

Faraday's Law	$\nabla \times \mathbf{E}(\mathbf{r},t) = -\frac{\partial \mathbf{B}(\mathbf{r},t)}{\partial t} - \mathbf{M}(\mathbf{r},t)$
Ampere's Law	$\nabla \times \mathbf{H}(\mathbf{r},t) = \frac{\partial \mathbf{D}(\mathbf{r},t)}{\partial t} + \mathbf{J}(\mathbf{r},t)$
Gauss's Law	$\nabla \cdot \mathbf{D}(\mathbf{r},t) = \rho_e(\mathbf{r},t)$
	$\nabla \cdot \mathbf{B}(\mathbf{r},t) = \rho_m(\mathbf{r},t)$



FDTD

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

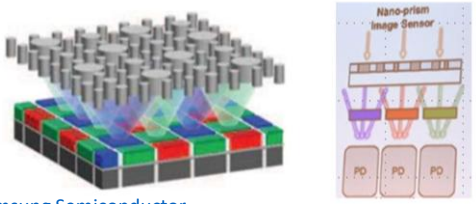
$$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}\right)$$

FEM

$$\nabla \times \nabla \times \mathbf{E}$$

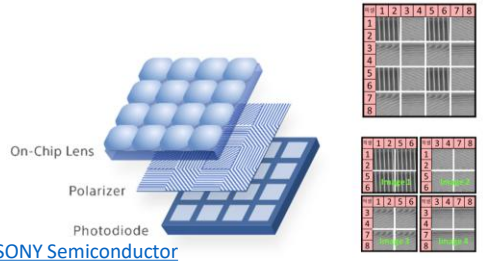
$$\nabla \times \nabla \times \mathbf{B}$$

Type of Information



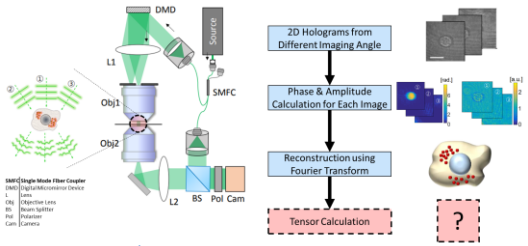
Samsung Semiconductor

Intensity of Wavelength



SONY Semiconductor

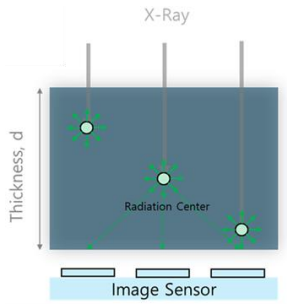
Intensity of Polarization

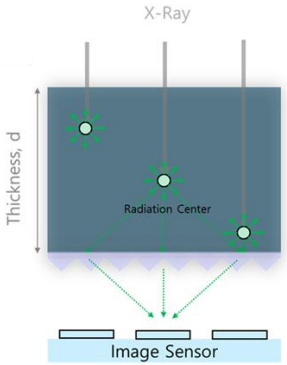


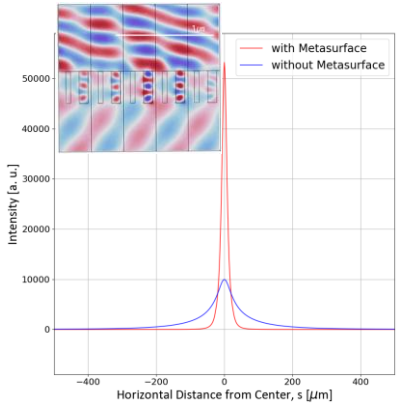
HT-X1 - Tomocube, Inc.

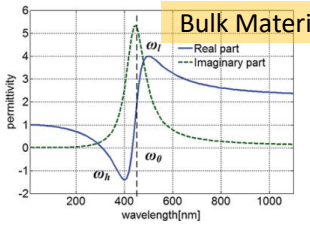
Intensity of Phase

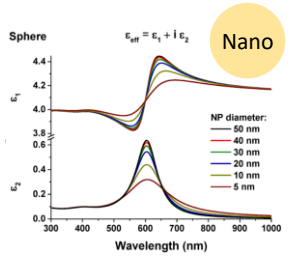
Metasurface*





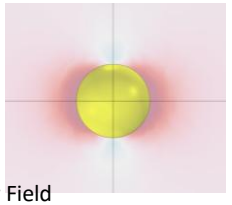




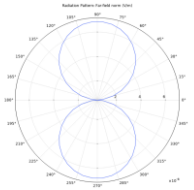


Helmholtz Equation	$(\nabla^2 + k^2)\varphi(r) = -\frac{\rho(r)}{\varepsilon_0 \varepsilon_m}$
Electrodynamic Potential	$\varphi(r) = \frac{1}{4\pi\varepsilon_0\varepsilon_m} \int_{V'} d^3r' \rho(\mathbf{r}') \frac{e^{jk\lvert\mathbf{r}-\mathbf{r}'\rvert}}{\lvert\mathbf{r}-\mathbf{r}'\rvert}$
Numerator	$e^{jk\lvert\mathbf{r}-\mathbf{r}'\rvert} \underset{Taylor}{\sim} e^{jkr}\{1 - jk\bar{\mathbf{a}}_\mathbf{r} \cdot \mathbf{r}' + \cdots\}$
Denominator	$\frac{1}{\lvert\mathbf{r}-\mathbf{r}'\rvert} \underset{Taylor}{\sim} \frac{1}{r}\left\{1 + \frac{r'}{r}\cos\theta - \frac{1}{2}\left(\frac{r'}{r}\right)^2 + \cdots\right\}$
Dipole Radiation	$\mathbf{E}(r) = -\nabla\varphi(r) - \frac{\partial \mathbf{A}}{\partial t}$

Analytic Calculation



Near Field



Far Field

Numerical Calculation

Plasmonics**

* S Kim et al., Negative Refraction Dielectric Metasurface for High Spatial Resolution X-Ray Detector, Korea Photonic Conference (2025): W2C-V.05
** S Kim, K Yoon, "Method and apparatus for measuring surface-enhanced raman scattering", WO2025170222A1

Thanks